UNITED ARAB EMIRATES MINISTRY OF EDUCATION
الإمارات العربية المتحدة وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 12 Physics Final Exam Practice

Term 1 2026–2027 • Set C • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. A 2.2 kg book is raised from 0.80 m to 2.10 m. Its potential-energy increase is: [4 marks]

A. 28.0 J
B. 45.3 J
C. 2.86 J
D. 17.2 J

Reference: p. 20, Practice Problem 32 | **Answer A**

2. Which expression gives the rest energy associated with mass m? [4 marks]

A. E = mc
B. E = ½mc
C. E = m/c²
D. E = mc²

Reference: pp. 20–22 | **Answer D**

3. For an isolated elastic collision, which quantities are conserved? [4 marks]

A. Momentum only, with kinetic energy always lost
B. Kinetic energy only
C. Neither momentum nor kinetic energy
D. Momentum and total kinetic energy

Reference: pp. 29–31 | **Answer D**

4. Which proposal resolved the classical problem of blackbody radiation? [4 marks]

A. Energy is emitted in every arbitrarily small amount equally
B. Energy exchange occurs in discrete quanta
C. Temperature does not affect radiation
D. All objects emit one identical frequency

Reference: pp. 46–48 | **Answer B**

5. In Planck's model an oscillator's allowed energies are given by: [4 marks]

A. $E_n = n/hf$
B. $E_n = nhf$
C. $E_n = hf/n$ for every state
D. $E_n = nf/h$

Reference: p. 48 | **Answer B**

6. One electron volt is the energy change of an electron moved through a potential difference of: [4 marks]

A. 1 Hz
B. 1 N
C. 1 V
D. 1 A

Reference: pp. 49–51 | **Answer C**

7. For light of wavelength 650 nm, photon energy is approximately: [4 marks]

A. 0.524 eV
B. 1.91 eV
C. 6.50 eV
D. 8.06×10^5 eV

Reference: p. 58, Check Your Progress 21 | **Answer B**

8. On a graph of KE_{max} versus incident frequency, the slope is: [2 marks]

A. The electron mass
B. The work function φ
C. The speed of light
D. Planck's constant h

Reference: pp. 53–55 | **Answer D**

9. The Compton effect supports the idea that light: [4 marks]

A. Cannot interact with electrons
B. Has no momentum
C. Transfers momentum in particle-like collisions
D. Always retains its original wavelength in collisions

Reference: p. 58, Check Your Progress 24 | **Answer C**

10. If a particle's momentum doubles, its de Broglie wavelength: [4 marks]

A. Doubles
B. Halves
C. Quadruples
D. Stays unchanged

Reference: pp. 59–60 | **Answer B**

11. The uncertainty principle is best described as: [4 marks]

A. A method for measuring charge
B. An error that disappears with perfect rulers
C. A fundamental limit, rather than merely poor instrument design
D. Proof that electrons have no momentum

Reference: p. 61 | **Answer C**

12. Thomson's atomic model pictured electrons: [4 marks]

A. Orbiting a tiny nucleus on quantized paths
B. Inside a neutron-only nucleus
C. With no positive charge anywhere
D. Embedded in a spread-out positive charge

Reference: pp. 62–63 | **Answer D**

13. A dark absorption line occurs when atoms absorb photons whose energy: [4 marks]

A. Matches an allowed upward energy transition
B. Has any arbitrary value equally
C. Must exceed the entire rest energy of the atom
D. Is always zero

Reference: pp. 64–66 | **Answer A**

14. For hydrogen, which transition requires absorption? [4 marks]

A. n = 1 to n = 2
B. n = 4 to n = 1
C. n = 3 to n = 2
D. n = 2 to n = 1

Reference: p. 72, Example Problem 4 | **Answer A**

15. As n increases in a hydrogen atom, its allowed energy: [4 marks]

A. Approaches zero from below
B. Is always positive
C. Becomes increasingly negative without limit
D. Remains −13.6 eV

Reference: pp. 69–71 | **Answer A**

16. The quantum mechanical model describes an atomic electron using: [2 marks]

A. An orbit independent of its energy state
B. A stationary point in the nucleus
C. A probability distribution
D. One precisely known classical circular path at all times

Reference: pp. 75–79 | **Answer C**

Set C Section B Free Response

17. A spring exerts an average force of 2.0 N on a 0.080 kg uniform solid pinball over 5.0 cm. The ball starts from rest and leaves rolling without slipping on a level surface. Use $I = \frac{2}{5}mr^2$ and $\omega = v/r$. [10 marks]

Reference: p. 28, Practice Problem 43; pp. 23–25

(a) Calculate the work done by the spring. [2]

Answer $W = F_{average}\,\Delta x$ [1] = 2.0(0.050) = 0.100 J [1].

(b) Express total kinetic energy in terms of m and v. [3]

Answer $KE_{total} = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$ [1]; substitute I and ω [1]; $KE_{total} = 0.70mv^2$ [1].

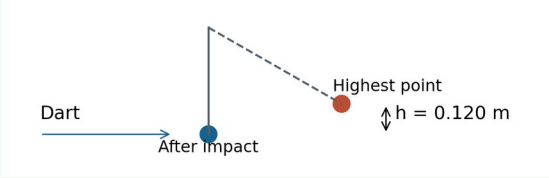
(c) Find the ball's speed. [3]

Answer $0.100 = 0.70(0.080)v^2$ [1]; $v^2 = 1.786$ [1]; v = 1.34 m/s [1].

(d) Calculate translational and rotational kinetic energies. [2]

Answer $KE_{trans} = 0.0714$ J [1]; $KE_{rot} = 0.0286$ J [1].

18. A 0.025 kg magnetic dart embeds in a stationary 0.73 kg suspended target. The combined objects then swing upward by 0.120 m before stopping momentarily. Ignore air resistance and losses during the swing. [10 marks]



Reference: p. 32, Practice Problem 46

(a) Distinguish the conservation principle used during impact from the one used during the swing. [2]

Answer During impact, horizontal momentum is conserved and KE is not [1]; during the swing, mechanical energy is conserved [1].

(b) Calculate the common speed just after impact. [3]

Answer $\frac{1}{2}Mv^2 = Mgh$ [1]; $v = \sqrt{(2 \times 9.8 \times 0.120)}$ [1] = 1.53 m/s [1].

(c) Calculate the initial dart speed. [3]

Answer $m_d u = (m_d + m_t)v$ [1]; $u = (0.755/0.025)(1.5336)$ [1] = 46.3 m/s [1].

(d) Calculate the kinetic energy just after impact and explain why it is less than before. [2]

Answer $KE_{after} = Mgh = 0.888$ J [1]; impact converts part of initial KE to internal energy/deformation [1].

19. A metal illuminated with 193 nm ultraviolet radiation emits photoelectrons with maximum kinetic energy 3.5 eV. Use hc = 1240 eV nm and h = 4.14 × 10⁻¹⁵ eV s. [10 marks]

Reference: p. 56, Practice Problem 14; p. 55, Figure 9

(a) Find the incident photon energy and the metal's work function. [3]

Answer $E = 1240/193 = 6.425$ eV [1]; $\varphi = E - KE_{max}$ [1] = 2.925 eV [1].

(b) Find threshold frequency and threshold wavelength. [3]

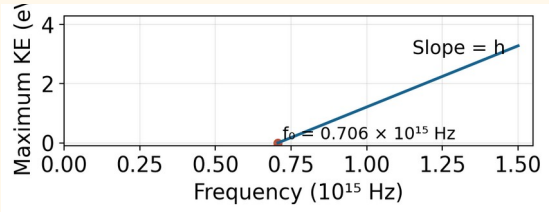
Answer $f_0 = \varphi/h$ [1] = 7.07×10^{14} Hz [1]; $\lambda_0 = 1240/\varphi$ = 424 nm [1].

(c) Sketch KE_{max} versus frequency; label f_0 and indicate how the work function is obtained. [2]

Answer Correct labelled increasing line through $(f_0,0)$ [1]; $\varphi = hf_0$, or magnitude of extrapolated energy-axis intercept [1].

(d) Would 532 nm light eject photoelectrons from this metal? Explain. [2]

Answer No [1]; 1240/532 = 2.33 eV < 2.925 eV, below threshold [1].



20. For a mercury transition, the atomic energy drops from 8.82 eV to 6.67 eV. Separately, a helium ion has ground-state energy −54.4 eV; a transition to that state emits a 30.4 nm photon. Use hc = 1240 eV nm and h = 4.14 × 10⁻¹⁵ eV s. [10 marks]

Reference: p. 73, Practice Problems 38–39; pp. 64–69

(a) Calculate the mercury photon's energy and wavelength. [3]

Answer $E_{photon} = 8.82 - 6.67 = 2.15$ eV [1]; $\lambda = hc/E$ [1] = 577 nm [1].

(b) Calculate the mercury photon's frequency and draw a labelled downward transition. [3]

Answer $f = E/h$ [1] = 5.19×10^{14} Hz [1]; labelled 8.82 and 6.67 eV levels and downward arrow [1].

(c) Find the helium-ion photon energy and the initial excited-state energy. [3]

Answer $E_{photon} = 1240/30.4 = 40.8$ eV [1]; $E_{initial} - (-54.4) = 40.8$ [1]; $E_{initial} = -13.6$ eV [1].

(d) Explain why atomic spectra contain discrete lines. [1]

Answer Only specific energy-level differences, and therefore specific photon energies/wavelengths, are allowed [1].

